

Side Effects

How Drug Causes Side Effects

- Theoretically due to downstream effects of blocking orexin receptors

Notable Side Effects

- Sedation, headache, abnormal dreams



Life-Threatening or Dangerous Side Effects

- Sleep paralysis, hypnagogic/hypnopompic hallucinations, and cataplexy-like symptoms (rare)

Weight Gain



- Reported but not expected

Sedation



- Many experience and/or can be significant in amount

What to Do About Side Effects

- Wait
- To avoid problems with memory, take lemborexant only if planning to have a full night's sleep

- Lower the dose
- Switch to a different hypnotic

Best Augmenting Agents for Side Effects

- Many side effects cannot be improved with an augmenting agent

Dosing and Use

Usual Dosage Range

- 5 mg/night

Dosage Forms

- Tablet 5 mg, 10 mg

How to Dose

- Starting dose is 5 mg, no more than once per night immediately before going to bed
- Should not take unless there are at least 7 hours remaining of sleep time



Dosing Tips

- Patients who tolerate but do not respond to 5 mg may receive the 10 mg dose; 10 mg is the maximum recommended dose
- Use the lowest effective dose for the patient

- Taking lemborexant with or soon after a meal can delay the time to sleep onset
- Not restricted to short-term use

Overdose

- Limited data; sedation

Long-Term Use

- Has been evaluated and found effective in trials up to 1 year

Habit Forming

- Lemborexant is under review by the DEA for scheduling
- No evidence of physiological dependence or withdrawal symptoms in clinical trials

How to Stop

- Taper not necessary

Pharmacokinetics

- Terminal elimination half-life 52.8 hours; effective half-life 17–29 hours
- Metabolized primarily by CYP450 3A4 and to a lesser extent by CYP450 3A5

- Absorption is delayed in the presence of food

Drug Interactions

- Not recommended in patients taking concomitant strong or moderate CYP450 3A4 inhibitors
- Patients taking mild CYP450 3A4 inhibitors should receive a maximum dose of 5 mg/night
- Not recommended in patients taking concomitant strong or moderate CYP450 3A4 inducers
- Lemborexant decreases the AUC of drugs that are CYP450 2B6 substrates (e.g., bupropion, methadone), potentially reducing their effectiveness
- Increased depressive effects when taken with other CNS depressants

Other Warnings/Precautions

- Insomnia may be a symptom of a primary disorder, rather than a primary disorder itself
- Some patients may exhibit abnormal thinking or behavioral changes similar to those caused by other CNS depressants (i.e., either depressant actions or disinhibiting actions)
- Some depressed patients may experience a worsening of suicidal ideation
- Use only with caution in patients with impaired respiratory function or obstructive sleep apnea

- Lemborexant should only be administered at bedtime
- Label contains warning for risk of next-day impaired alertness and motor coordination
- Sleep driving and other complex behaviors, such as eating and preparing food and making phone calls, have been reported in patients taking sedative hypnotics; if complex sleep behaviors occur, discontinue use

Do Not Use

- If patient has narcolepsy
- If there is a proven allergy to lemborexant

Special Populations

Renal Impairment

- Dose adjustment not necessary
- Somnolence may be increased in patients with severe renal impairment

Hepatic Impairment

- Dose adjustment not necessary for mild hepatic impairment
- Initial and maximum recommended dose 5 mg/night for moderate hepatic impairment

- Not recommended for patients with severe hepatic impairment

Cardiac Impairment

- Not studied in patients with cardiac impairment

Elderly

- Some patients may tolerate lower doses better

Children and Adolescents

- Safety and efficacy have not been established

Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- When administered during organogenesis, the no observed adverse effect levels (NOAELs) for fetal toxicity are approximately >100 and 23 times the maximum recommended human dose (MRHD) based on AUC in rats and rabbits, respectively

- When administered during pregnancy and lactation, the NOAEL for toxicity is 93 times the MRHD based on AUC in rats
- Pregnancy registry for lemborexant: 1-888-274-2378

Breast Feeding

- Unknown if lemborexant is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- Recommended either to discontinue drug or bottle feed

The Art of Psychopharmacology

Potential Advantages

- Primary insomnia
- Chronic insomnia
- Those who require long-term treatment
- Those with depression whose insomnia does not resolve with antidepressant treatment

Potential Disadvantages

- More expensive than some other sedative hypnotics

Primary Target Symptoms

- Time to sleep onset

- Nighttime awakenings
- Total sleep time

Pearls

- DORAs are reversible inhibitors of orexin, so their action can theoretically be reversed as endogenous orexin is released upon awakening, and even to a certain extent in the middle of the night
- This is unlike Z-drug and benzodiazepine hypnotics, which do not reverse upon wakening
- DORAs hypothetically switch off alertness-related circuits, countering inappropriate wakefulness associated with insomnia, whereas Z-drug and benzodiazepine hypnotics and melatonin hypothetically switch on sleep-related circuits
- Curbing alertness-related factors in patients with insomnia may have some theoretical efficacy and tolerability advantages over enhancing sleep drives, sedation, and somnolence
- Targeting insomnia may prevent the onset of major depressive disorder (MDD) or generalized anxiety disorder (GAD) and help maintain remission after recovery from MDD or GAD
- Rebound insomnia was not observed in clinical trials
- May be effective in patients with insomnia unresponsive to medications with other mechanisms of action (e.g., Z-drug hypnotics, benzodiazepines)

Suggested Reading

Murphy P , Moline M , Mayleben D , et al. Lemborexant, a dual orexin receptor antagonist (DORA) for the treatment of insomnia disorder: results from a Bayesian, adaptive, randomized, double-blind, placebo-controlled study . *J Clin Sleep Med* 2017 ;**13** (11):1289–99 .

Rosenberg R , Murphy P , Zammit G , et al. Comparison of lemborexant with placebo and zolpidem tartrate extended release for the treatment of older adults with insomnia disorder: a phase 3 randomized clinical trial . *JAMA Netw Open* 2019 ;**2** (12):e1918254 . doi: 10.1001/jamanetworkopen.2019.18254 .

Vermeeren A , Jongen S , Murphy M , et al. On-the-road driving performance the morning after bedtime administration of lemborexant in healthy adult and elderly volunteers . *Sleep* 2019 ;**42** (4). pii: zsy260. doi: 10.1093/sleep/zsy260 .

Levetiracetam

Therapeutics

Brands

- Keppra
- Keppra XR
- *see index for additional brand names*

Generic?

Yes

Class

- Anticonvulsant, synaptic vesicle protein SV2A modulator

Commonly Prescribed for

(bold for FDA approved)

- **Adjunct therapy for partial seizures in patients with epilepsy (>16 years of age for extended-release, >4 years of age for immediate-release)**
- **Adjunct therapy for myoclonic seizures in juvenile myoclonic epilepsy (ages 12 and older)**

- **Adjunct therapy for primary generalized tonic–clonic seizures in idiopathic generalized epilepsy (ages 6 and older)**
- Neuropathic pain/chronic pain
- Mania

How the Drug Works

- * Binds to synaptic vesicle protein SV2A, which is involved in synaptic vesicle exocytosis
- Opposes the activity of negative modulators of GABA- and glycine-gated currents and partially inhibits N-type calcium currents in neuronal cells

How Long Until It Works

- Should reduce seizures by 2 weeks
- Not yet clear if it has mood-stabilizing effects in bipolar disorder or antineuralgic actions in chronic neuropathic pain, but some patients may respond and if so, would be expected to show clinical effects starting by 2 weeks although it may take several weeks to months to optimize clinical effects

If It Works

- The goal of treatment is complete remission of symptoms (e.g., seizures, mania, pain)

- The goal of treatment of chronic neuropathic pain is to reduce symptoms as much as possible, especially in combination with other treatments
- Treatment of chronic neuropathic pain most often reduces but does not eliminate symptoms and is not a cure since symptoms usually recur after medicine stopped
- Continue treatment until all symptoms are gone or until mood is stable and then continue treating indefinitely as long as improvement persists
- Continue treatment indefinitely to avoid recurrence of seizures, mania, and pain

If It Doesn't Work (for Bipolar Disorder or Neuropathic Pain)

- * May be effective only in a subset of bipolar patients, in some patients who fail to respond to other mood stabilizers, or it may not work at all
- Many patients have only a partial response where some symptoms are improved but others persist or continue to wax and wane without stabilization of pain or mood
- Other patients may be nonresponders, sometimes called treatment-resistant or treatment-refractory
- Consider increasing dose or switching to another agent with better demonstrated efficacy in bipolar disorder or neuropathic pain



Best Augmenting Combos for Partial Response or Treatment Resistance

- Levetiracetam is itself a second-line augmenting agent to numerous other anticonvulsants, lithium, and atypical antipsychotics for bipolar disorder and to gabapentin, tiagabine, other anticonvulsants, SNRIs, and TCAs for neuropathic pain

Tests

- None for healthy individuals

Side Effects

How Drug Causes Side Effects

- CNS side effects may be due to excessive actions on SV2A synaptic vesicle proteins or to actions on various voltage-sensitive ion channels

Notable Side Effects

- * Sedation, dizziness, ataxia, asthenia
- Hematologic abnormalities (decrease in red blood cell count and hemoglobin)



Life-Threatening or Dangerous Side Effects

- Rare severe dermatologic reactions [Stevens-Johnson syndrome (SJS), toxic epidermal necrolysis (TEN)]

- Activation of suicidal ideation and acts (rare)
- Changes in behavior (aggression, agitation, anxiety, hostility)
- Rare activation of suicidal ideation and behavior (suicidality)

Weight Gain



- Reported but not expected

Sedation



- Many experience and/or can be significant in amount

What to Do About Side Effects

- Wait
- Wait
- Wait
- Take more of the dose at night to reduce daytime sedation
- Lower the dose
- Switch to another agent

Best Augmenting Agents for Side Effects

- Many side effects cannot be improved with an augmenting agent

Dosing and Use

Usual Dosage Range

- 1000–3000 mg/day in 2 doses

Dosage Forms

- Tablet 250 mg, 500 mg, 750 mg, 1000 mg
- Extended-release tablet 500 mg, 750 mg, 1000 mg, 1500 mg
- Tablet for suspension 250 mg, 500 mg, 750 mg, 1000 mg
- Oral solution 100 mg/mL
- Intravenous injection 500 mg/5 mL
- Intravenous injection (levetiracetam in sodium chloride) 500 mg/100 mL, 1000 mg/100 mL, 1500 mg/100 mL

How to Dose

- Initial 1000 mg/day in 1 (extended-release) or 2 (immediate-release) doses; after 2 weeks can increase by 1000 mg/day every 2 weeks; maximum dose generally 3000 mg/day



Dosing Tips

- For intolerable sedation, can give most of the dose at night and less during the day
- Some patients may tolerate and respond to doses greater than 3000 mg/day

Overdose

- No fatalities; sedation, agitation, aggression, respiratory depression, coma

Long-Term Use

- Safe

Habit Forming

- No

How to Stop

- Taper
- Epilepsy patients may seize upon withdrawal, especially if withdrawal is abrupt
- * Rapid discontinuation can increase the risk of relapse in bipolar disorder
- Discontinuation symptoms uncommon

Pharmacokinetics

- Elimination half-life approximately 6–8 hours
- Inactive metabolites
- Not metabolized by CYP450 enzymes
- Does not inhibit/induce CYP450 enzymes

- Renally excreted



Drug Interactions

- Because levetiracetam is not metabolized by CYP450 enzymes and does not inhibit or induce CYP450 enzymes, it is unlikely to have significant pharmacokinetic drug interactions



Other Warnings/Precautions

- Depressive effects may be increased by other CNS depressants (alcohol, MAOIs, other anticonvulsants, etc.)
- Whenever possible, warn patients and their caregivers about the possibility of activating side effects, and advise them to report such side effects immediately
- Monitor patients for behavioral symptoms (aggression, agitation, anger, anxiety, apathy, depression, hostility, irritability) as well as for possible psychotic symptoms or suicidality

Do Not Use

- If there is a proven allergy to levetiracetam

Special Populations

Renal Impairment

- Recommended dose for patients with mild impairment may be between 500 mg and 1500 mg twice a day
- Recommended dose for patients with moderate impairment may be between 250 mg and 750 mg twice a day
- Recommended dose for patients with severe impairment may be between 250 mg and 500 mg twice a day
- Patients on dialysis may require doses between 500 mg and 1000 mg once a day, with a supplemental dose of 250–500 mg following dialysis

Hepatic Impairment

- Dose adjustment usually not necessary

Cardiac Impairment

- No specific recommendations

Elderly

- Some patients may tolerate lower doses better
- Elderly patients may be more susceptible to adverse effects



Children and Adolescents

- Safety and efficacy not established under age 16

- Children may require higher doses than adults; dosing should be adjusted according to weight

Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- Use in women of childbearing potential requires weighing potential benefits to the mother against the risks to the fetus
- Antiepileptic Drug Pregnancy Registry: 1-888-233-2334, www.aedpregnancyregistry.org
- Taper drug if discontinuing
- Seizures, even mild seizures, may cause harm to the embryo/fetus
- Lack of convincing efficacy for treatment of bipolar disorder or chronic neuropathic pain suggests risk/benefit ratio is in favor of discontinuing levetiracetam during pregnancy for these indications
- * For bipolar patients, given the risk of relapse in the postpartum period, mood-stabilizer treatment, especially with agents with better evidence of efficacy than levetiracetam, should generally be restarted immediately after delivery if patient is unmedicated during pregnancy

- * For bipolar patients, levetiracetam should generally be discontinued before anticipated pregnancies
- * Atypical antipsychotics may be preferable to levetiracetam if treatment of bipolar disorder is required during pregnancy
- Bipolar symptoms may recur or worsen during pregnancy and some form of treatment may be necessary

Breast Feeding

- Some drug is found in mother's breast milk
- * Recommended either to discontinue drug or bottle feed
- If drug is continued while breast feeding, infant should be monitored for possible adverse effects
- If infant becomes irritable or sedated, breast feeding or drug may need to be discontinued
- * Bipolar disorder may recur during the postpartum period, particularly if there is a history of prior postpartum episodes of either depression or psychosis
- * Relapse rates may be lower in women who receive prophylactic treatment for postpartum episodes of bipolar disorder
- Atypical antipsychotics and anticonvulsants such as valproate may be safer than levetiracetam during the postpartum period when breast feeding

The Art of Psychopharmacology

Potential Advantages

- Patients on concomitant drugs (lack of drug interactions)
- Treatment-refractory bipolar disorder
- Treatment-refractory neuropathic pain

Potential Disadvantages

- Patients noncompliant with twice daily dosing
- Efficacy for bipolar disorder or neuropathic pain not well documented

Primary Target Symptoms

- Seizures
- Pain
- Mania



Pearls

- Well studied in epilepsy
- * Off-label use second-line and as an augmenting agent may be justified for bipolar disorder and neuropathic pain unresponsive to other treatments

- * Unique mechanism of action suggests utility where other anticonvulsants fail to work
- * Unique mechanism of action as modulator of synaptic vesicle release suggests theoretical utility for clinical conditions that are hypothetically linked to excessively activated neuronal circuits, such as anxiety disorders and neuropathic pain as well as epilepsy

Suggested Reading

Ben-Menachem E . Levetiracetam: treatment in epilepsy . *Expert Opin Pharmacother* 2003 ;**4** (11):2079 –88.

French J . Use of levetiracetam in special populations . *Epilepsia* 2001 ;**42** (Suppl 4):S40 –3.

Leppik IE . Three new drugs for epilepsy: levetiracetam, oxcarbazepine, and zonisamide . *J Child Neurol* 2002 ;**17** (Suppl 1):S53 –7.

Lynch BA , Lambeng N , Nocka K , et al. The synaptic vesicle protein SV2A is the binding site for the antiepileptic drug levetiracetam . *Proc Natl Acad Sci USA* 2004 ;**101** :9861 –6.

Pinto A , Sander JW . Levetiracetam: a new therapeutic option for refractory epilepsy . *Int J Clin Pract* 2003 ;**57** (7):616–21 .

Levomilnacipran

Therapeutics

Brands

- Fetzima
- *see index for additional brand names*

Generic?

No

Class

- Neuroscience-based Nomenclature: serotonin, norepinephrine reuptake inhibitor (SN-RI)
- SNRI (dual serotonin and norepinephrine reuptake inhibitor); antidepressant

Commonly Prescribed for

(bold for FDA approved)

- **Major depressive disorder**
- Fibromyalgia
- Neuropathic pain/chronic pain

How the Drug Works

- Boosts neurotransmitters serotonin, norepinephrine/noradrenaline, and dopamine
- Blocks norepinephrine reuptake pump (norepinephrine transporter), presumably increasing noradrenergic neurotransmission
- Blocks serotonin reuptake pump (serotonin transporter), presumably increasing serotonergic neurotransmission
- Presumably desensitizes both serotonin 1A receptors and beta adrenergic receptors
- Since dopamine is inactivated by norepinephrine reuptake in frontal cortex, which largely lacks dopamine transporters, levomilnacipran can increase dopamine neurotransmission in this part of the brain

How Long Until It Works

- Onset of therapeutic actions is usually not immediate, but often delayed 2–4 weeks
- If it is not working within 6 or 8 weeks, it may require a dosage increase (off label), or it may not work at all
- May continue to work for many years to prevent relapse of symptoms

If It Works

- The goal of treatment of depression is complete remission of current symptoms as well as prevention of future relapses

- Treatment of depression most often reduces or even eliminates symptoms, but not a cure since symptoms can recur after medicine stopped
- Continue treatment of depression until all symptoms are gone (remission) or significantly reduced
- Once symptoms are gone, continue treating for 1 year for the first episode of depression
- For second and subsequent episodes of depression, treatment may need to be indefinite

If It Doesn't Work

- Many depressed patients only have a partial response where some symptoms are improved but others persist (especially insomnia, fatigue, and problems concentrating)
- Other depressed patients may be nonresponders, sometimes called treatment-resistant or treatment-refractory
- Some depressed patients who have an initial response may relapse even though they continue treatment, sometimes called “poop-out”
- Consider increasing dose, switching to another agent, or adding an appropriate augmenting agent
- Consider psychotherapy
- Consider evaluation for another diagnosis or for a comorbid condition (e.g., medical illness, substance abuse, etc.)